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Inventor(s)

Giladi; Alexander et al.

METHODS, SYSTEMS, AND APPARATUSES TO MITIGATE SERVER-ASSOCIATED DELAYS IN CONTENT DELIVERY

Abstract

Client devices in a content delivery network may send requests for content to upstream devices along with one or more parameters that indicate, as an example, client-device-related conditions or constraints, such as available buffer space or buffer starvation time. These parameters may be used by the upstream devices to prioritize content requests according to a prioritization schedule to ensure requested content is timely delivered to the appropriate client device(s) at the appropriate time to prevent playback failure, stalls, etc.

Inventors: Giladi; Alexander (Centennial, CO), Balk; Alexander (Centennial, CO), Begen; Ali C. (Centennial, CO), Benton; Johnathon (Centennial, CO)

Applicant: Comcast Cable Communications, LLC (Philadelphia, PA)

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Background/Summary

BACKGROUND

[0001] Content servers may cache content items to ensure timely delivery to requesting client devices. When a requested content item is not presently cached at the content server that receives the request, the content item may need to be retrieved from another content server, which may introduce delays in fulfilling the corresponding content request. These and other considerations are discussed herein.

SUMMARY

[0002] It is to be understood that both the following general description and the following detailed description are exemplary and explanatory only and are not restrictive. This summary is not intended to identify critical or essential features, but merely to summarize certain features and variations. Client devices in a content delivery network (“CDN”) may send requests for content to upstream devices, such as content servers, along with one or more parameters that indicate client-device-related conditions or constraints. For example, a client device may indicate an available buffer space, a buffer starvation time, etc., via one or more parameters sent with a request.

[0003] Such parameters sent with content requests may be used by an upstream device(s) to prioritize responses to those content requests according to a prioritization schedule. As an example, when requested content is not available at an upstream device closest to an “edge” of the CDN relative to a requesting client device (e.g., when a “cache miss” occurs at the edge), the content may be requested from another upstream device closer to the “top” of the CDN, such as a mid-tier upstream device for example. However, doing so may introduce delays in fulfilling the content request, and such delays may be exacerbated when a same upstream device receives multiple content requests from multiple client devices that may each indicate differing conditions or constraints. To mitigate such delays, responses to content requests may be sent according to the prioritization schedule to ensure requested content is timely delivered to the appropriate client device(s) at the appropriate time (e.g., to prevent or mitigate playback failure, stalls, etc.) Other details and features will be described in the sections that follow.

Description

BRIEF DESCRIPTION OF THE DRAWINGS

[0004] The accompanying drawings, which are incorporated in and constitute a part of this specification, together with the description, serve to explain the principles of the present methods and systems:

[0005] FIG. 1 shows an example system;

[0006] FIG. 2A shows an example workflow for content delivery;

[0007] FIG. 2B shows an example workflow for content delivery;

[0008] FIG. 3 shows an example system;

[0009] FIG. 4 shows a flowchart for an example method;

[0010] FIG. 5 shows a flowchart for an example method; and

[0011] FIG. 6 shows a flowchart for an example method.

DETAILED DESCRIPTION

[0012] As used in the specification and the appended claims, the singular forms “a,” “an,” and “the” include plural referents unless the context clearly dictates otherwise. Ranges may be expressed herein as from “about” one particular value, and/or to “about” another particular value. When such a range is expressed, another configuration includes from the one particular value and/or to the other particular value. When values are expressed as approximations, by use of the antecedent “about,” it will be understood that the particular value forms another configuration. It will be further understood that the endpoints of each of the ranges are significant both in relation to the other endpoint, and independently of the other endpoint.

[0013] “Optional” or “optionally” means that the subsequently described event or circumstance may or may not occur, and that the description includes cases where said event or circumstance occurs and cases where it does not.

[0014] Throughout the description and claims of this specification, the word “comprise” and variations of the word, such as “comprising” and “comprises,” means “including but not limited to,” and is not intended to exclude other components, integers, or steps. “Exemplary” means “an example of” and is not intended to convey an indication of a preferred or ideal configuration. “Such as” is not used in a restrictive sense, but for explanatory purposes.

[0015] It is understood that when combinations, subsets, interactions, groups, etc. of components are described that, while specific reference of each various individual and collective combinations and permutations of these may not be explicitly described, each is specifically contemplated and described herein. This applies to all parts of this application including, but not limited to, steps in described methods. Thus, if there are a variety of additional steps that may be performed it is understood that each of these additional steps may be performed with any specific configuration or combination of configurations of the described methods.

[0016] As will be appreciated by one skilled in the art, hardware, software, or a combination of software and hardware may be implemented. Furthermore, a computer program product on a computer-readable storage medium (e.g., non-transitory) having processor-executable instructions (e.g., computer software) embodied in the storage medium. Any suitable computer-readable storage medium may be utilized including hard disks, CD-ROMs, optical storage devices, magnetic storage devices, memristors, Non-Volatile Random Access Memory (NVRAM), flash memory, or a combination thereof.

[0017] Throughout this application, reference is made to block diagrams and flowcharts. It will be understood that each block of the block diagrams and flowcharts, and combinations of blocks in the block diagrams and flowcharts, respectively, may be implemented by processor-executable instructions. These processor-executable instructions may be loaded onto a general-purpose computer, special purpose computer, or other programmable data processing apparatus to produce a machine, such that the processor-executable instructions which execute on the computer or other programmable data processing apparatus create a device for implementing the functions specified in the flowchart block or blocks.

[0018] These processor-executable instructions may also be stored in a computer-readable memory that may direct a computer or other programmable data processing apparatus to function in a particular manner, such that the processor-executable instructions stored in the computer-readable memory produce an article of manufacture including processor-executable instructions for implementing the function specified in the flowchart block or blocks. The processor-executable instructions may also be loaded onto a computer or other programmable data processing apparatus to cause a series of operational steps to be performed on the computer or other programmable apparatus to produce a computer-implemented process such that the processor-executable instructions that execute on the computer or other programmable apparatus provide steps for implementing the functions specified in the flowchart block or blocks.

[0019] Accordingly, blocks of the block diagrams and flowcharts support combinations of devices for performing the specified functions, combinations of steps for performing the specified functions and program instruction means for performing the specified functions. It will also be understood that each block of the block diagrams and flowcharts, and combinations of blocks in the block diagrams and flowcharts, may be implemented by special purpose hardware-based computer systems that perform the specified functions or steps, or combinations of special purpose hardware and computer instructions.

[0020] “Content items,” as the phrase is used herein, may also be referred to as “content,” “content data,” “content information,” “content asset,” “multimedia asset data file,” or simply “data” or “information”. Content items may be any information or data that may be licensed to one or more

individuals (or other entities, such as business or group). Content may be electronic representations of video, audio, text, and/or graphics, which may be but is not limited to electronic representations of videos, movies, or other multimedia, which may be but is not limited to data files adhering to H.264/MPEG-AVC, H.265/MPEG-HEVC, H.266/MPEG-VVC, MPEG-5 EVC, MPEG-5 LCEVC, AV1, MPEG2, MPEG, MPEG4 UHD, SDR, HDR, 4k, Adobe® Flash® Video (.FLV), ITU-T H.261, ITU-T H.262 (MPEG-2 video), ITU-T H.263, ITU-T H.264 (MPEG-4 AVC), ITU-T H.265 (MPEG HEVC), ITU-T H.266 (MPEG VVC) or some other video file format, whether such format is presently known or developed in the future. The content items described herein may be electronic representations of music, spoken words, or other audio, which may be but is not limited to data files adhering to MPEG-1 audio, MPEG-2 audio, MPEG-2 and MPEG-4 advanced audio coding, MPEG-H, AC-3 (Dolby Digital), E-AC-3 (Dolby Digital Plus), AC-4, Dolby Atmos®, DTS®, and/or any other format configured to store electronic audio, whether such format is presently known or developed in the future. Content items may be any combination of the above-described formats.

[0021] “Consuming content” or the “consumption of content,” as those phrases are used herein, may also be referred to as “accessing” content, “providing” content, “viewing” content, “listening” to content, “rendering” content, or “playing” content, among other things. In some cases, the particular term utilized may be dependent on the context in which it is used. Consuming video may also be referred to as viewing or playing the video. Consuming audio may also be referred to as listening to or playing the audio. This detailed description may refer to a given entity performing some action. It should be understood that this language may in some cases mean that a system (e.g., a computer) owned and/or controlled by the given entity is actually performing the action.

[0022] FIG. 1 shows an example system **100** for content delivery. The system **100** may comprise a plurality of computing devices/entities in communication via a network **110**. The network **110** may be an optical fiber network, a coaxial cable network, a hybrid fiber-coaxial network, a wireless network, a satellite system, a direct broadcast system, an Ethernet network, a high-definition multimedia interface network, a Universal Serial Bus (USB) network, or any combination thereof. Data may be sent on the network **110** via a variety of transmission paths, including wireless paths (e.g., satellite paths, Wi-Fi paths, cellular paths, etc.) and terrestrial paths (e.g., wired paths, a direct feed source via a direct line, etc.). The network **110** may comprise public networks, private networks, wide area networks (e.g., Internet), local area networks, and/or the like. The network **110** may comprise a content access network, content distribution network, and/or the like. The network **110** may be configured to provide content from a variety of sources using a variety of network paths, protocols, devices, and/or the like. The content delivery network and/or content access network may be managed (e.g., deployed, serviced) by a content provider, a service provider, and/or the like. The network **110** may deliver content items from a source(s) to a user device(s).

[0023] The system **100** may comprise a source **102**, such as a server or other computing device. The source **102** may receive source streams for a plurality of content items. The source streams may be live streams (e.g., a linear content stream) and/or video-on-demand (VOD) streams. The live streams may comprise, for example, low-latency (“LL”) live streams. The source **102** may receive the source streams from an external server or device (e.g., a stream capture source, a data storage device, a media server, etc.). The source **102** may receive the source streams via a wired or wireless network connection, such as the network **110** or another network (not shown).

[0024] The source **102** may comprise a headend, a video-on-demand server, a cable modem termination system, and/or the like. The source **102** may provide content (e.g., video, audio, games, applications, data) and/or content items (e.g., video, streaming content, movies, shows/programs, etc.) to user devices. The source **102** may provide streaming media, such as live content, on-demand content (e.g., video-on-demand), content recordings, and/or the like. The source **102** may be managed by third-party content providers, service providers, online content providers, over-the-top content providers, and/or the like. A content item may be provided via a subscription, by

individual item purchase or rental, and/or the like. The source **102** may be configured to provide content items via the network **110**. Content items may be accessed by user devices via applications, such as mobile applications, television applications, set-top box applications, gaming device applications, and/or the like. An application may be a custom application (e.g., by a content provider, for a specific device), a general content browser (e.g., a web browser), an electronic program guide, and/or the like.

[0025] The source **102** may provide uncompressed content items, such as raw video data, comprising one or more portions (e.g., frames/slices, groups of pictures (GOP), coding units (CU), coding tree units (CTU), etc.). It should be noted that although a single source **102** is shown in FIG. **1**, this is not to be considered limiting. In accordance with the described techniques, the system **100** may comprise a plurality of sources **102**, each of which may receive any number of source streams.

[0026] The system **100** may comprise an encoder **104**, such as a video encoder, a content encoder, etc. The encoder **104** may be configured to encode one or more source streams received via the source **102** into a plurality of content items/streams at various bitrates (e.g., various representations/quality levels). For example, the encoder **104** may be configured to encode a source stream for a content item at varying bitrates for corresponding representations (e.g., versions/quality levels) of a content item for adaptive bitrate streaming. As shown in FIG. **1**, the encoder **104** may encode a source stream into Representations 1-5. It is to be understood that the FIG. **1** shows five representations for explanation purposes only. The encoder **104** may be configured to encode a source stream into fewer or greater representations. Representation 1 may be associated with a first resolution (e.g., 480p) and/or a first bitrate (e.g., 2 Mbps). Representation 2 may be associated with a second resolution (e.g., 720p) and/or a second bitrate (e.g., 3.5 Mbps). Representation 3 may be associated with a third resolution (e.g., 1080p) and/or a third bitrate (e.g., 6 Mbps). Representation 4 may be associated with a fourth resolution (e.g., 4K) and/or a first bitrate (e.g., 18 Mbps). Representation 5 may be associated with a fifth resolution (e.g., 8K) and/or a fifth bitrate (e.g., 45 Mbps). Other example resolutions and/or bitrates are possible. Note that “Representation” is a term defined for MPEG DASH (ISO/IEC 23009-1), while Apple HTTP Live Streaming (IETF RFC 8216) (hereinafter, “HLS”) defines the same concept as a “variant,” and the present methods, systems, and apparatuses are not intended to be limited to DASH-based environments/use cases.

[0027] The system **100** may comprise a packager **106**. The packager **106** may be configured to receive one or more content items/streams from the encoder **104**. The packager **106** may be configured to prepare content items/streams for distribution. For example, the packager **106** may be configured to convert encoded content items/streams into a plurality of content fragments. The packager **106** may be configured to provide content items/streams according to adaptive bitrate streaming. For example, the packager **106** may be configured to convert encoded content items/streams at various representations into one or more adaptive bitrate streaming formats, such as Apple HTTP Live Streaming (HLS), Microsoft Smooth Streaming, Adobe HTTP Dynamic Streaming (HDS), MPEG DASH, any media streaming format based on the TCP or reliable UDP (e.g., Quick UDP Internet Connections, “QUIC”) transport protocol, driven by client requests, and/or the like. The packager **106** may pre-package content items/streams and/or provide packaging in real-time as content items/streams are requested by user devices, such as a user device **112** and a user device **113**. The user devices **112** and **113** may each be a content/media player, a set-top box, a client device, a smart device, a mobile device, a user device, etc. Though only two user devices are shown in FIG. **1**, it is to be understood that the system **100** may comprise fewer or greater user devices.

[0028] The system **100** may comprise a content server **108**. The content server **108** may be configured to receive requests for content, such as content items/streams. The content server **108** may identify a location of a requested content item and provide the content item—or a portion thereof—to a device requesting the content, such as the user device **112** and/or the user device **113**.

The content server **108** may be configured to provide a communication session with a requesting device, such as the user device **112**, based on HTTP, FTP, or other protocols. The content server **108** may be one of a plurality of content servers distributed across the system **100**. The content server **108** may be located in a region proximate to the user device **112**. A request for a content stream/item from the user device **112** may be directed to the content server **108** (e.g., due to the location and/or network conditions). The content server **108** may be configured to deliver content streams/items to the user device **112** in a specific format requested by the user device **112**. The content server **108** may be configured to provide the user device **112** with a manifest file (e.g., or other index file describing portions of the content) corresponding to a content stream/item. The content server **108** may be configured to provide streaming content (e.g., unicast, multicast) to the user device **112**. The content server **108** may be configured to provide a file transfer and/or the like to the user device **112**. The content server **108** may cache or otherwise store content (e.g., frequently requested content) to enable faster delivery of content items to users. The content server **108** may receive a request for a content item, such as a request for high-resolution video and/or the like. The content server **108** may receive the request for the content item from the user device **112**. As further described herein, the content server **108** may be capable of sending (e.g., to the user device **112**) one or more portions of the content item at varying bitrates (e.g., Representations 1-5). [0029] The system **100** may comprise a content server **109** that provides similar functionality as the content server **108**. The content server **109** may be “upstream” with respect to the content server **108** and/or the user devices **112,113**. For example, the content server **109** may be “closer” in terms of network hops to an origin/source of the content relative to the content server **108**. Additionally, or in the alternative, the content server **109** may comprise—or be a part of—a content origin(s), a mezzanine feed(s), etc. Though only two content servers are shown in FIG. **1**, it is to be understood that the system **100** may comprise fewer or greater content servers.

[0030] The system **100** may be configured to mitigate server-associated delays in content delivery that cause latency, which may lead to re-buffering at the client devices (e.g., the user device **112** and/or **113**). And cache misses at content servers at the edge (e.g., the content server **108** and/or **109**) may add noticeable latency. A main reason for latency is that a request may need to be sent to another content server (e.g., a higher-level content server) or/and an origin server (e.g., the source **102**) when a cache miss(es) occurs in order for the content server at the edge to respond to the corresponding content request(s). This situation puts client devices with low buffer levels at a higher risk, as cache miss latency may result in re-buffering (e.g., due to empty buffer) or a shift to a lower-quality representation (e.g., if the buffer levels are low but sustainably so). Given increased probability of cache misses, the problem is more pronounced when requested content has low concurrency (e.g., a relatively small number of client devices are requesting the content) and/or when the content server at the edge does not serve a large number of clients.

[0031] The system **100** may mitigate such delays/latency by processing content requests according to a prioritization schedule. For example, the system **100** may process content requests according to a prioritization schedule that may be based on the prioritization scheme defined by the IETF in RFC 9218 (e.g., an HTTP/2 or HTTP/3-based prioritization scheme). However, it is to be understood that the prioritization scheme defined by the IETF in RFC 9218 is provided herein as merely one example of many examples the system **100** may use.

[0032] The prioritization scheme may be associated with one or more prioritization parameters. For example, the system **100** may use a first prioritization parameter, *u* (referred to herein as an “urgency parameter”), and a second prioritization parameter, *i* (referred to herein as an “incremental parameter”). The prioritization parameters may be communicated by client devices to content servers. For example, the prioritization parameters may be communicated by client devices to content servers via an HTTP Priority header field and/or a PRIORITY_UPDATE frame. The HTTP Priority header field and/or the PRIORITY_UPDATE frame may be carried within (e.g., sent via) an HTTP/3 Control Stream. For example, a request indicative of the urgency parameter

having a value of $u=3$ may represent an urgency level of “3” (e.g., on a scale from 0-7, with 0 being the highest urgency level), and a request indicative of the incremental parameter i being “yes” or “true” may indicate the response to be provided may be sent incrementally (e.g., the response may be sent in parts). Examples of incremental responses (e.g., responses that may be sent in parts) include streaming content/media, like videos or audio (e.g., streamed in chunks, allowing for continuous playback as more data is received). Non-incremental responses, on the other hand, may be those that must be fully received before they may be used, executed, parsed, etc. Examples of non-incremental responses may include manifest files, configuration files, etc. (e.g., data, files, etc., that generally must be fully downloaded and available before they may be opened, parsed, or executed).

[0033] Each of the content servers **108**, **109** may communicate with another content server(s) (e.g., a higher-level content server(s)) and/or an origin server(s) (e.g., the source **102**) when a cache miss(es) occurs. And the corresponding content request(s) associated with the cache miss(es) may be multiplexed on a same session or connection or stream with the other content server(s). For example, the corresponding content request(s) may be sent from the content server experiencing the cache miss(es) to the other content server (e.g., a higher-level content server) via a same session or connection or stream between the two servers. For example, the corresponding content request(s) associated with the cache miss(es) may be multiplexed on (e.g., sent via) a same HTTP/2 connection/session/stream, a same HTTP/3 connection/session/stream, a same Quick UDP Internet Connection (“QUIC”) connection/session/stream, and/or the like rather than via separate connections/sessions. In other words, multiple content requests may be sent from a first content server(s) (e.g., a lower-level content server(s)) to another content server(s) (e.g., a higher-level content server(s)) and/or an origin server(s) via a same session, connection, etc., versus using a separate connection/session/stream per request. And, as further described herein, the other content server(s) and/or the origin server(s) may respond to a multiplexed content request(s) based on (e.g., according to) a prioritization schedule, which itself may be based on client-device-related constraints or conditions indicated by the requesting client device(s). For purposes of explanation, the description herein is written as communication between a content server at the edge (e.g., the content server **108**) and a higher-level content server (e.g., the source **102** and/or the content server **109**); however, the description herein may apply equally to any two content servers having a parent-child relationship and logically (e.g., in terms of a network, such as the network **110**) between a client device(s) and an origin server (e.g., a content source, such as the source **102**).

[0034] As one example, the user devices **112,113** may be geographically closest to (and/or a least number of network hops away from) the content server **108** versus the content server **109** (e.g., the content server **108** may be at the “edge” of the system **100** relative to the user devices **112,113**). The content server **108** may receive separate content requests, which may or may not be associated with a same portion of content or content item, from the user devices **112,113**. However, the requested content may not be present in cache/storage associated with the content server **108** (e.g., the content server **108** may experience a “cache miss” with respect to the content requests). The content server **108** may be at a “lower-level” in the system's **100** network hierarchy (e.g., at a mid-tier) with respect to the content server **109**, which may be—in at least this example—at a “higher-level” than the content server **108** (e.g., at a higher tier). In response to the content server **108** determining that the requested content is not present in cache/storage (e.g., in response to the content server **108** experiencing a “cache miss”), the content server **108** may retrieve/request the requested content from, in at least this example, the content server **109**. However, instead of sending separate requests to the content server **109** that would each use a separate connection/session/stream, the content server **108** may send a multiplexed request to the content server **109** via a same session or connection or stream between the two servers. The content server **108** may send the multiplexed request to the content server **109** by sending separate requests (e.g., based on each request received from the user devices **112,113**) on/via a same session or connection

or stream with the content server **109**. For example, the corresponding content request(s) may be sent from the content server **108** to the content server **109** via a same HTTP/2 connection/session/stream, a same HTTP/3 connection/session/stream, a same Quick UDP Internet Connection (“QUIC”) connection/session/stream, and/or the like, rather than via separate connections/sessions/streams.

[0035] As described herein, the system **100** may process content requests according to a prioritization schedule, which may indicate values for a first prioritization parameter, *u* (the “urgency parameter”), and a second prioritization parameter, *i* (the “incremental parameter”) for each associated content request. The system **100** may comprise an upstream request scheduler (hereinafter, a “scheduler”, not shown). The scheduler may modify values of the aforementioned prioritization parameters indicated by content requests (e.g., the urgency parameter and the incremental parameter). For example, the scheduler may modify a request indicative of an urgency parameter having a value of *u*=3 (an urgency level of “3”) to a greater or lesser value, depending on various factors discussed herein. The scheduler may maintain a prioritization schedule to track each content request received and modifications that may be made to the prioritization parameters. The prioritization schedule may comprise, for example, a priority queue, a table, a hash table, a linked list, a custom data structure, a combination thereof, and/or the like.

[0036] The scheduler and/or the prioritization schedule may be resident at each of the content servers **108** and **109** of the system **100**. Additionally, or in the alternative, the scheduler and/or the prioritization schedule may be resident at another device within the system **100** (not shown) that is in communication with each of the content servers **108** and **109**. The scheduler may assign (e.g., modify/reassign) higher priorities according to high-priority client devices (e.g., the user device **112** and/or **113**), high-priority content, higher-priority request types, a combination thereof, and/or the like. There may also be classes of client devices deemed high priority due to business rules (e.g., content output at certain establishments, such as those showing sporting events, should not encounter stalls or delays for business reasons). Additionally, a same session (e.g., an HTTP/2 or HTTP/3 connection/session, a Quick UDP Internet Connections (“QUIC”) connection/session, etc.) may be used for prefetching content resources that are highly likely to be requested, especially if associated with high-priority content, such as initialization segments, manifest files, etc.

[0037] A high-priority client device (e.g., the user device **112** and/or **113**) may be a “struggling client device” at risk of quality degradation or re-buffering due to low buffer level. For example, the client device may indicate a low buffer condition to the content server using the Common Media Client Data specification as defined by the Consumer Technology Association; however, other examples for communicating the low buffer condition are possible as well. A high-priority client device may also be a client device outputting content at very low latency and/or at slower playback speeds in order to avoid stalling. For example, a client device may comprise a playback buffer for storing content that is to be output (e.g., played, displayed, etc.) at a later time, and the content request may comprise an indication of a status of the playback buffer (a “buffer status”).

[0038] A client device's buffer status may be indicative of one or more of: a buffer length, a buffer size, an amount of time, an amount of memory, a combination thereof, and/or the like. Additionally, or in the alternative, the buffer status may comprise and/or indicate a buffer starvation time and/or a buffer length parameter. The buffer length parameter may represent and/or indicate a size of content stored in the buffer (e.g., memory size) and/or a length of content stored in the buffer (e.g., an amount of time). The content server (e.g., the content server **108** and/or **109**) may determine the buffer starvation time based on the buffer status indicated. For example, the content server may determine the buffer starvation time based on the buffer length parameter, and the buffer starvation time may comprise an amount of time that the client device may output one or more portions of the content that are presently stored in the playback buffer (e.g., an amount of time until the playback buffer becomes depleted). The content server that receives the corresponding content request from the requesting client device may prioritize the request in correlation with a shallowness of the client

device's buffer (e.g., based on the buffer status). For example, the content server may prioritize the request in correlation with the shallowness of the client device's buffer when the client device's buffer is shorter than some threshold, T (e.g., a period of time, amount of time, etc.),

[0039] Examples of content that the scheduler may determine to be high-priority content may include low-latency linear content or/and high-concurrency content likely to generate a large number of requests for the same content segment(s) from multiple client devices. High-priority request types may as content requests where the requested content has a high impact on quality of experience (e.g., content quality upon playback/output). For example, a content request may be assigned a high-priority request type by the scheduler when the requested content is a manifest file (e.g., a DASH MPD) and/or an initialization segment(s), since both are smaller than content segments and both are essential for starting video playback.

[0040] As another example, a higher priority may be assigned to a content request when the requested content is a lowest-rate/lowest-quality version available and/or the requested content is audio content (e.g., an audio track(s)). The lowest-rate/lowest-quality version is typically requested by either struggling client devices (e.g., struggling to start or continue playback) and/or by client devices that are building-up their buffer to start playback. Audio content, such as one or more audio tracks, may be assigned a higher priority because a struggling client device may be able to at least continue audio playback even when corresponding video playback stalls or fails. Additionally, if scalable coding is used, enhancement layer segments may be prioritized lower.

[0041] In low-latency live streaming, a single content segment may comprise a small number of frames (e.g., 100 milliseconds worth of frames, or even a single frame), and such content segments are referred to as “partial segments,” as they are not necessarily independently playable and are solely intended to reduce transmission delay. For example, if the requested content is a partial segment that does not carry any independently-playable frames (e.g., “IDR” frames, etc.), then the corresponding content request may be assigned a lower priority by the scheduler, while partial segments with independently-playable frames may be assigned a higher priority by the scheduler. When low-delay streaming is being used, special representations intended for fast start-up (e.g., as defined in SCTE **214-6** or DASH-IF IOP) may also be prioritized by the scheduler.

[0042] The scheduler, which again may be resident at the content server **108** and/or **109**, may further reprioritize a given content request based on its deadline. The deadline may be computed given data points such as buffer duration at the client device (e.g., the user device **112** and/or **113**), round trip time, link throughput, a size (real or estimated) of the response, a combination thereof, and/or the like. In this way, content requests that are “in flight” (e.g., in-process) may be prioritized higher as they get closer to their deadline. This requires the scheduler to periodically re-assess the progress of the ongoing content requests.

[0043] In some examples, a struggling client device may cancel a content request for a content segment at a high or a medium bitrate/quality level and subsequently send a content request for the same content segment at a lower bitrate/quality level. In such examples, the scheduler and/or the corresponding content server may not cancel the request for the content segment at the high or the medium bitrate/quality level but may rather deprioritize it while still fulfilling the subsequently-sent content request for the same content segment at the lower bitrate/quality level.

[0044] In some examples, the content request may be for a manifest file (e.g., a DASH MPD), and the content server that receives the request may determine a cache miss (e.g., the manifest file is not available at the content server), which may cause the content request to be forwarded to an origin server (e.g., content source). In such examples, the scheduler and/or the origin server may add an additional Early Hint response (e.g., an HTTP status of **103**) listing URLs for content resources that are likely to be requested next. For example, the URLs listed may be associated with an initialization segment associated with the requested manifest file, an XLink response(s), a subset of content segments associated with the requested manifest file that are either more likely to be requested next and/or are the lowest-rate/lowest-quality versions of those content segments (e.g.,

those that may be provided in order to avoid playback stalling or failure), a combination thereof, and/or the like. The scheduler and/or the origin server may then determine which of the aforementioned content resources may result in a cache miss if requested by the client device and may further request those content resources, or a subset thereof, based on the content server's possible knowledge of bitrate(s)/quality level(s) requested.

[0045] In examples where low-latency live streaming is being used, the content server may request a manifest file or/and a content segment(s) that is not yet available (e.g., not yet generated, packaged, and/or encoded). The content server may request the manifest file or/and the content segment(s) that is not yet available in response to the corresponding client device sending a “blocking request” as defined in Apple Low Latency-HLS and/or a miscalculated timing (e.g., via CMCD to indicate a next requested object), for example. In this case, the request may be made with low priority and the scheduler may reprioritize fulfillment of the request once the requested object, or a portion thereof, becomes available.

[0046] The scheduler may receive indications from client devices related to network congestion, such as “explicit congestion notifications” (ECN) indicated by content requests. Additionally, or in the alternative, the scheduler, which may be resident at a content server closest to the “edge” (e.g., closest to the requesting client device), may respond to duress signals sent by another content server (e.g. via CMSD). For example, the scheduler may cause prefetching requests to be paused or/and dropped. As another example, the scheduler may deprioritize requests for content segments at a highest-available bitrate/quality level, especially for low-priority content. For example, client devices streaming content at 4K resolution may be downgraded to 1440p and later to 1080p. In such examples, the scheduler and/or the content server closest to the edge may indicate a maximum allowed bandwidth and/or the duress signals to the corresponding client device (e.g., using the Common Media Server Data specification, as defined by the Consumer Technology Association). Additionally, or in the alternative, if layered coding such as LC-EVC, S-HEVC, or SVC is used, enhancement layer segments may be deprioritized or dropped.

[0047] In some examples, there may be dependencies, either explicit or implicit, between content requests (e.g., requests for portions of the same content item). For example, when a content request received from a client device (e.g., the user device **112** and/or **113**), the request may have an associated “deadline” that may represent when the client device “needs” the requested content/portion, such as to avoid playback stalls when an associated buffer is almost depleted. If such a deadline is missed for a requested base layer segment when scalable coding is used, for example, a deadline for an enhancement layer segment may be disregarded, since use of the enhancement layer segment is dependent upon availability of the corresponding base layer segment. Similarly, if a requested low-latency (LL) partial segment/portion of content carrying an independently-playable frame(s) failed its corresponding “deadline,” then any associated deadline(s) for any following portion(s) of the content that depends on the independently-playable frame(s) for decoding may be extended. These examples, as well as other possible examples for mitigating server-associated delays, are further illustrated in the description herein for example workflows **200A** and **200B** shown in FIGS. **2A** and **2B**, as well as methods **400-600** shown in FIGS. **4-6** and further described herein.

[0048] As an example, FIG. **2A** shows the example workflow **200A**, which may be implemented by the system **100** when a cache miss occurs. At step **202A**, the user device **112** may send a request for content. The user device **112** may send the request to the content server **108** directly. Additionally, or in the alternative, the request may be sent from the user device **112** to one or more intermediary devices/components of the system **100** (e.g., servers, caches, etc.—not shown in FIG. **1**), which may send (e.g., route, forward, etc.) the request to the content server **108**. The request may comprise any suitable message for requesting the content, such as a request for a segment of the content, a chunk of the content, a manifest (or portion thereof) for the content, a combination thereof, and/or the like.

[0049] The content server **108** may receive the request. Based on the request, the content server **108** may determine whether the content is available locally. For example, the content server **108** may determine whether the corresponding segment, chunk, and/or manifest for the content is available at a cache(s) of the content server **108** or at a storage repository readily accessible by the content server **108** (e.g., within a same network, a same server group, etc.). The content server **108** may determine that the content requested by the user device **112** is not locally available. Such a scenario may be referred to herein as a “cache miss.” When the content server **108** determines the cache miss (e.g., determines the unavailability of the content locally), the content server **108** may request and/or retrieve the content from another device/component of the system **100**. The content server **108** may determine which server, cache, or storage repository of the system **100** has the content available based on caching records, caching rules, caching schedules, load balancing rules, content delivery rules, a combination thereof, and/or the like. For example, the content server **108** may determine that the content is available—or the content server **108** may simply inquire whether the content is available—at the content server **109**.

[0050] At step **204A**, the content server **108** may request and/or retrieve the content from the content server **109**. For example, the content server **108** may request and/or retrieve the corresponding segment, chunk, and/or manifest (or portion thereof) for the content from the content server **109**. The content server **108** may send a request for the content to the content server **109** directly or via one or more intermediary devices/components of the system **100** (e.g., servers, caches, etc.—not shown in FIG. 1), which may send (e.g., route, forward, etc.) the request to the content server **109**. The request sent to the content server **109** at step **204A** may be a multiplexed request associated with a plurality of content requests (e.g., received from the user device **112**, the user device **113**, and/or other user/client devices not shown). For example, the content server **108** may send the multiplexed request to the content server **109** by sending separate requests (e.g., based on each content request received) on/via a same session or connection or stream with the content server **109**, such as a same HTTP/2 connection/session/stream, a same HTTP/3 connection/session/stream, a same Quick UDP Internet Connection (“QUIC”) connection/session/stream, and/or the like, rather than via separate connections/sessions/streams. The content server **109** may receive the multiplexed request inasmuch the content server **109** may receive each of the separate requests sent by the content server **108** via the same HTTP/3 connection/session/stream, the same QUIC connection/session/stream, and/or the like, rather than via separate connections/sessions/streams.

[0051] The multiplexed request may be indicative of, and/or comprise, the prioritization schedule described herein. For example, each content request of the plurality of content requests may indicate at least one prioritization parameter associated with a corresponding user/client device that sent the corresponding content request to the content server **108**, and the prioritization schedule may indicate the at least one prioritization parameter for each content request of the plurality of content requests. The content server **109** may process each of the content requests according to the prioritization schedule. For example, a content request associated with the multiplexed request may be processed before another content request that was also received as part of the multiplexed request if the at least one prioritization parameter (e.g., urgency parameter) for the content request indicates a buffer starvation time and/or a buffer length parameter that is less than a buffer starvation time and/or a buffer length parameter indicated by the at least one prioritization parameter (e.g., urgency parameter) for the other content request. Other examples are possible as well (e.g., based on an incremental parameter, etc.) At step **206A**, the content server **109** may send the content to the content server **108**. For example, the content server **109** may send the corresponding segment, chunk, and/or manifest (or portion thereof) for the content to the content server **108**. The content server **108** may receive the content from the content server **109**. At step **208A**, the content server **108** may send the content to the user device **112**.

[0052] FIG. 2B shows the example workflow **200B**. The workflow **200B** may be implemented by

the system **100** as part of the workflow **200A** or it may be implemented separately. For example, the workflow **200B** may be implemented by the system **100** when requested content is not immediately provided to a requesting device/component (e.g., a client/user device) due to the prioritization schedule of the system **100** discussed herein.

[0053] At step **202B**, the user device **112** may send a request for content. The user device **112** may send the request to the content server **108** directly. Additionally, or in the alternative, the request may be sent from the user device **112** to one or more intermediary devices/components of the system **100** (e.g., servers, caches, etc.—not shown in FIG. **1**), which may send (e.g., route, forward, etc.) the request to the content server **108**. The request may comprise any suitable message for requesting the content, such as a request for a segment of the content, a chunk of the content, a manifest (or portion thereof) for the content, a combination thereof, and/or the like. The user device **112** may comprise a playback buffer for storing content that is to be output (e.g., played, displayed, etc.) at a later time, and the request sent by the user device **112** may comprise an indication of a status of the playback buffer. For example, the indication of the status of the playback buffer may be communicated by the user device **112** via an urgency parameter included with the request, as described herein. The status of the playback buffer may comprise and/or indicate (e.g., via the urgency parameter) a buffer starvation and/or a buffer length parameter. The buffer length parameter may represent and/or indicate a size of content stored in the buffer (e.g., memory size) and/or a length of content stored in the buffer (e.g., an amount of time). The prioritization schedule may be implemented by the system **100** to prevent the buffer becoming depleted and causing a stall in content output. For example, the user device **112** may encounter a stall when a next portion(s) of content being output is not received in a timely manner (e.g., prior to content in the buffer being output).

[0054] The content server **108** may determine a buffer starvation time based on the status of the playback buffer (e.g., based on the urgency parameter). For example, the content server **108** may determine the buffer starvation time based on the buffer length parameter indicated by the urgency parameter sent with the request. The buffer starvation time may comprise an amount of time that the user device **112** may output one or more portions of the content that are presently stored in the playback buffer (e.g., an amount of time until the playback buffer becomes depleted).

[0055] At step **204B**, the content server **108** (e.g., via the scheduler) may determine that the request sent by the user device **112** is to be placed in a queue of requests (e.g., not immediately fulfilled). The content server **108** may place the request in the queue based on the prioritization schedule. For example, the request may indicate the prioritization parameter (e.g., $u=3$) having an urgency level of “3”, and the content server **108** may modify/reassign the urgency level indicated by the request. For example, the content server **108** may determine that the request received by the user device **112** at step **202B** is to be processed after another request sent by the user device **113** (not shown). The other request sent by the user device **113** may comprise a prioritization parameter (e.g., an urgency parameter) indicating a status of a playback buffer of the user device **113**, such as a corresponding buffer length parameter. The prioritization parameter sent by the user device **113** may indicate a buffer starvation time for the user device **113** that is smaller (e.g., earlier in time) than the buffer starvation time for the user device **112**. The scheduler may cause the content server **108** to prioritize the request sent by the user device **113** over the request sent by the user device **112**. For example, the content server **108** may cause the urgency level indicated by the request sent by the user device **113** to be modified to a lower/more urgent level and/or placed ahead in the queue, etc., such that the request sent by the user device **113** may be processed before the request sent by the user device **112** is processed.

[0056] At step **206B**, the content server **108** (e.g., via the scheduler) may process the request sent by the user device **112** (e.g., after processing the request sent by the user device **113**). It is to be understood that the scheduler may cause the content server **108** (and/or the content server **109**) to prioritize one request over another for other reasons as well. For example, the scheduler may cause

the content server **108** (and/or the content server **109**) to prioritize a request based on content type, request type, client type, content popularity, device type, device class, service/subscriber type or level, a combination thereof, and/or the like.

[0057] The present methods and systems may be computer-implemented. FIG. 3 shows a block diagram depicting a system/environment **300** comprising non-limiting examples of a computing device **301** and a server **302** connected through a network **304**. Either of the computing device **301** or the server **302** may be a computing device, such as any of the devices of the system **100** shown in FIG. 1. In an aspect, some or all steps of any described method may be performed on a computing device as described herein. The computing device **301** may comprise one or multiple computers configured to store parameter data **329** (e.g., relating to prioritization parameters, prioritization schedule(s), etc.), and/or the like. The server **302** may comprise one or multiple computers configured to store content data **324** (e.g., a plurality of content segments, parameters, etc.). Multiple servers **302** may communicate with the computing device **301** via the through the network **304**.

[0058] The computing device **301** and the server **302** may be a digital computer that, in terms of hardware architecture, generally includes a processor **308**, system memory **310**, input/output (I/O) interfaces **312**, and network interfaces **314**. These components (**308**, **310**, **312**, and **314**) are communicatively coupled via a local interface **316**. The local interface **316** may be, for example, but not limited to, one or more buses or other wired or wireless connections, as is known in the art. The local interface **316** may have additional elements, which are omitted for simplicity, such as controllers, buffers (caches), drivers, repeaters, and receivers, to enable communications. Further, the local interface may include address, control, and/or data connections to enable appropriate communications among the aforementioned components.

[0059] The processor **308** may be a hardware device for executing software, particularly that stored in system memory **310**. The processor **308** may be any custom made or commercially available processor, a central processing unit (CPU), an auxiliary processor among several processors associated with the computing device **301** and the server **302**, a semiconductor-based microprocessor (in the form of a microchip or chip set), or generally any device for executing software instructions. When the computing device **301** and/or the server **302** is in operation, the processor **308** may execute software stored within the system memory **310**, to communicate data to and from the system memory **310**, and to generally control operations of the computing device **301** and the server **302** pursuant to the software.

[0060] The I/O interfaces **312** may be used to receive user input from, and/or for providing system output to, one or more devices or components. User input may be provided via, for example, a keyboard and/or a mouse. System output may be provided via a display device and a printer (not shown). I/O interfaces **312** may include, for example, a serial port, a parallel port, a Small Computer System Interface (SCSI), an infrared (IR) interface, a radio frequency (RF) interface, and/or a universal serial bus (USB) interface.

[0061] The network interface **314** may be used to transmit and receive from the computing device **301** and/or the server **302** on the network **304**. The network interface **314** may include, for example, a 10BaseT Ethernet Adaptor, a 10BaseT Ethernet Adaptor, a LAN PHY Ethernet Adaptor, a Token Ring Adaptor, a wireless network adapter (e.g., WiFi, cellular, satellite), or any other suitable network interface device. The network interface **314** may include address, control, and/or data connections to enable appropriate communications on the network **304**.

[0062] The system memory **310** may include any one or combination of volatile memory elements (e.g., random access memory (RAM, such as DRAM, SRAM, SDRAM, etc.)) and nonvolatile memory elements (e.g., ROM, hard drive, tape, CDROM, DVDROM, etc.). Moreover, the system memory **310** may incorporate electronic, magnetic, optical, and/or other types of storage media. Note that the system memory **310** may have a distributed architecture, where various components are situated remote from one another, but may be accessed by the processor **308**.

[0063] The software in system memory **310** may include one or more software programs, each of which comprises an ordered listing of executable instructions for implementing logical functions. In the example of FIG. **3**, the software in the system memory **310** of the computing device **301** may comprise the parameter data **329**, the content data **324**, and a suitable operating system (O/S) **318**. In the example of FIG. **3**, the software in the system memory **310** of the server **302** may comprise the parameter data **329**, the content data **324**, and a suitable operating system (O/S) **318**. The operating system **318** essentially controls the execution of other computer programs and provides scheduling, input-output control, file and data management, memory management, and communication control and related services.

[0064] For purposes of illustration, application programs and other executable program components such as the operating system **318** are shown herein as discrete blocks, although it is recognized that such programs and components may reside at various times in different storage components of the computing device **301** and/or the server **302**. An implementation of the system/environment **300** may be stored on or transmitted across some form of computer readable media. Any of the disclosed methods may be performed by computer readable instructions embodied on computer readable media. Computer readable media may be any available media that may be accessed by a computer. By way of example and not meant to be limiting, computer readable media may comprise “computer storage media” and “communications media.” “Computer storage media” may comprise volatile and non-volatile, removable and non-removable media implemented in any methods or technology for storage of information such as computer readable instructions, data structures, program modules, or other data. Exemplary computer storage media may comprise RAM, ROM, EEPROM, flash memory or other memory technology, CD-ROM, digital versatile disks (DVD) or other optical storage, magnetic cassettes, magnetic tape, magnetic disk storage or other magnetic storage devices, or any other medium which may be used to store the desired information and which may be accessed by a computer.

[0065] FIG. **4** shows a flowchart of an example method **400** to mitigate server-associated delays in content delivery. The method **400** may be performed in whole or in part by a single computing device, a plurality of computing devices, and the like. For example, the steps of the method **400** may be performed by the content server **108**, the content server **109**, and/or a computing device in communication with the content server **108** or the content server **109**. Some steps of the method **400** may be performed by a first computing device (e.g., the content server **108**), while other steps of the method **400** may be performed by another computing device.

[0066] At step **410**, a first computing device (e.g., the content server **108**) may receive a plurality of content requests. The plurality of content requests may be associated with a plurality of client devices (e.g., the user devices **112,113**). At step **420**, the first computing device may determine, for each content request of the plurality of content requests, at least one prioritization parameter. For example, the first computing device may determine the at least one prioritization parameter for each content request based on (e.g., in response to) at least one portion of content associated with the corresponding content request being unavailable (e.g., due to a cache miss). The at least one prioritization parameter may comprise an urgency parameter and/or an incremental parameter.

[0067] In some examples, each content request, of the plurality of content requests, may be indicative of a buffer status associated with the client device corresponding to that content request. For example, when determining the at least one prioritization parameter for each content request of the plurality of content requests, the first computing device may determine the at least one prioritization parameter for each content request of the plurality of content requests based on the buffer status associated with the client device corresponding to the content request. The buffer status associated with a particular client device may be indicative of one or more of: a buffer length, a buffer size, an amount of time, an amount of memory, a combination thereof, and/or the like.

[0068] Additionally, or in the alternative, each content request, of the plurality of content requests,

may be indicative of a content type corresponding to that content request. The content type may comprise one or more of: low-latency live streaming content, low-delay streaming content, linear content, on-demand content, high-concurrency content, high-priority content, low-priority content, a combination thereof, and/or the like. The first computing device may determine the at least one prioritization parameter for each content request of the plurality of content requests based on the content type corresponding to the content request.

[0069] Additionally, or in the alternative, each content request, of the plurality of content requests, may be indicative of a request type corresponding to that content request. For example, the first computing device may determine the at least one prioritization parameter for each content request of the plurality of content requests based on the corresponding request type. The request type may comprise one or more of: a high-priority request, a low-priority request, a manifest request, a request for at least one initialization segment, a request for at least one partial segment, a request for at least one representation associated with low-delay streaming, a request for at least one enhancement layer segment, a combination thereof, and/or the like.

[0070] At step **430**, the first computing device may send a multiplexed request. The multiplexed request may be associated with the plurality of content requests. For example, the first computing device may send the multiplexed request to an upstream computing device (e.g., the content server **109** or the source **102**) by sending separate requests (e.g., based on each content request received) on/via a same session or connection or stream with the upstream computing device, such as a same HTTP/2 connection/session/stream, a same HTTP/3 connection/session/stream, a same Quick UDP Internet Connection (“QUIC”) connection/session/stream, and/or the like, rather than sending each content request via separate connections/sessions/streams. The upstream computing device may receive the multiplexed request inasmuch the upstream computing device may receive each of the separate content requests sent by the first computing device via the same HTTP/3 connection/session/stream, the same QUIC connection/session/stream, and/or the like, rather than via separate connections/sessions/streams. The multiplexed request may be indicative of, and/or comprise, the at least one prioritization parameter for each content request of the plurality of content requests.

[0071] At step **440**, the first computing device may receive the at least one portion of content for each content request of the plurality of content requests. The first computing device may receive the at least one portion of content for each content request of the plurality of content requests from the upstream computing device. The upstream computing device may determine a prioritization schedule associated with the plurality of content requests. For example, the upstream computing device may determine the prioritization schedule based on the multiplexed request and the at least one prioritization parameter for each content request of the plurality of content requests. The upstream computing device may send a plurality of responses to the first computing device based on the prioritization schedule. Each response of the plurality of responses may comprise the at least one portion of content for at least one content request of the plurality of content requests. At step **450**, the first computing device may cause the at least one portion of content associated with the content request corresponding to the client device to be output.

[0072] FIG. 5 shows a flowchart of an example method **500** to mitigate server-associated delays in content delivery. The method **500** may be performed in whole or in part by a single computing device, a plurality of computing devices, and the like. For example, the steps of the method **500** may be performed by the content server **108**, the content server **109**, and/or a computing device in communication with the content server **108** or the content server **109**. Some steps of the method **500** may be performed by a first computing device (e.g., the content server **108**), while other steps of the method **500** may be performed by another computing device.

[0073] At step **510**, a first computing device (e.g., the content server **109**) may receive a multiplexed request associated with a plurality of content requests. The plurality of content requests may each be associated with at least one portion of content. The first computing device may be

located “upstream” of a downstream computing device (e.g., a content server at the edge) closest to a plurality of client devices associated with the multiplexed request. The plurality of client devices may each send a content request to the downstream computing device, and the downstream computing device may send the multiplexed request to the first computing device by sending separate requests (e.g., based on each content request received) on/via a same session or connection or stream with the first computing device, such as a same HTTP/2 connection/session/stream, a same HTTP/3 connection/session/stream, a same Quick UDP Internet Connection (“QUIC”) connection/session/stream, and/or the like, rather than sending each content request via separate connections/sessions/streams. The first computing device may receive the multiplexed request inasmuch the first computing device may receive each of the separate content requests sent by the downstream computing device via the same HTTP/3 connection/session/stream, the same QUIC connection/session/stream, and/or the like, rather than via separate connections/sessions/streams. [0074] The downstream computing device may send the multiplexed request to the first computing device due to (e.g., responsive to) a cache miss(es) associated with the requested content. For example, the downstream computing device may receive the plurality of content requests, and, for each content request of the plurality of content requests, the downstream device may determine that the at least one portion of content associated with the content request is unavailable at the downstream device. The multiplexed request may be indicative of, and/or comprise, for each content request of the plurality of content requests, at least one prioritization parameter. The at least one prioritization parameter, for each content request of the plurality of content requests, may comprise an urgency parameter or an incremental parameter. And the multiplexed request may be indicative of, and/or comprise, the urgency parameter or the incremental parameter for each content request of the plurality of content requests.

[0075] Each content request of the plurality of content requests may indicate a buffer status associated with the client device. Additionally, or in the alternative, each content request may indicate a request type and/or a content type. The buffer status may be indicative of one or more of: a buffer length, a buffer size, an amount of time, an amount of memory, a combination thereof, and/or the like. The content type may comprise one or more of: low-latency live streaming content, low-delay streaming content, linear content, on-demand content, high-concurrency content, high-priority content, low-priority content, a combination thereof, and/or the like. The request type may comprise one or more of: a high-priority request, a low-priority request, a manifest request, a request for at least one initialization segment, a request for at least one partial segment, a request for at least one representation associated with low-delay streaming, a request for at least one enhancement layer segment, a combination thereof, and/or the like.

[0076] At step **520**, the first computing device may determine a prioritization schedule associated with the plurality of content requests. The first computing device may determine the prioritization schedule based on the at least one prioritization parameter for each content request of the plurality of content requests. The prioritization schedule may indicate, for each content request of the plurality of content requests, at least one of: a time or an order associated with sending the response to the downstream computing device. In some examples, based on at least one portion of content associated with the content request, at least one modification to the prioritization schedule may be determined. The at least one modification may be associated with a time or an order associated with sending at least one response associated with the at least one portion of content to the downstream computing device.

[0077] At step **530**, the first computing device may send a response, for each content request of the plurality of content requests, to the downstream computing device. Sending the responses to the downstream computing device may be based on the prioritization schedule. The responses may cause the at least one portion of content associated with the content request to be output at corresponding client devices associated with the plurality of content requests. Each response to each content request may comprise at least one portion of: a frame, a chunk, a segment, a manifest

file, a representation element associated with the at least one portion of content associated with the content request, a combination thereof, and/or the like.

[0078] FIG. 6 shows a flowchart of an example method **600** to mitigate server-associated delays in content delivery. The method **600** may be performed in whole or in part by a single computing device, a plurality of computing devices, and the like. For example, the steps of the method **600** may be performed by the content server **108**, the content server **109**, and/or a computing device in communication with the content server **108** or the content server **109**. Some steps of the method **600** may be performed by a first computing device (e.g., the content server **108**), while other steps of the method **600** may be performed by another computing device.

[0079] A first computing device (e.g., the content server **108**) may receive a plurality of content requests. Each content request of the plurality of content requests may indicate a buffer status associated with the client device. Additionally, or in the alternative, each content request may indicate a request type and/or a content type. The buffer status may be indicative of one or more of: a buffer length, a buffer size, an amount of time, an amount of memory, a combination thereof, and/or the like. The content type may comprise one or more of: low-latency live streaming content, low-delay streaming content, linear content, on-demand content, high-concurrency content, high-priority content, low-priority content, a combination thereof, and/or the like. The request type may comprise one or more of: a high-priority request, a low-priority request, a manifest request, a request for at least one initialization segment, a request for at least one partial segment, a request for at least one representation associated with low-delay streaming, a request for at least one enhancement layer segment, a combination thereof, and/or the like.

[0080] At step **610**, the first computing device may send a first multiplexed request. For example, the first computing device may send the first multiplexed request to a first upstream computing device (e.g., the content server **109**). The first multiplex request may be associated with a subset of a plurality of content requests. The first multiplexed request may be indicative of, and/or comprise, at least one prioritization parameter for each content request of the subset. For each content request of the subset, the first computing device may determine, based on the at least one portion of content associated with the content request being unavailable, the at least one prioritization parameter.

[0081] At step **620**, the first computing device may send a second multiplexed request. For example, the first computing device may send the second multiplexed request to a second upstream computing device (e.g., a content server upstream relative to the content server **108** and/or **109**). The second multiplex request may be associated with a remainder of the plurality of content requests. The second multiplexed request may be indicative of, and/or comprise, at least one prioritization parameter for each content request of the remainder. For each content request of the remainder, the first computing device may determine, based on the at least one portion of content associated with the content request being unavailable, the at least one prioritization parameter.

[0082] At step **630**, the first computing device may cause at least one portion of content for each content request of the subset to be output. For example, the first computing device may cause the at least one portion of content for each content request of the subset to be output according to a first prioritization schedule. The first prioritization schedule may be based on the at least one prioritization parameter for each content request of the subset.

[0083] At step **640**, the first computing device may cause, based on at least one portion of content for each content request of the remainder received via the second upstream computing device according to a second prioritization schedule, the at least one portion of content for each content request of the subset to be output by a remainder of the plurality of client devices. The second prioritization schedule may be based on the at least one prioritization parameter for each content request of the remainder.

[0084] While specific configurations have been described, it is not intended that the scope be limited to the particular configurations set forth, as the configurations herein are intended in all respects to be possible configurations rather than restrictive. Unless otherwise expressly stated, it is

in no way intended that any method set forth herein be construed as requiring that its steps be performed in a specific order. Accordingly, where a method claim does not actually recite an order to be followed by its steps or it is not otherwise specifically stated in the claims or descriptions that the steps are to be limited to a specific order, it is no way intended that an order be inferred, in any respect. This holds for any possible non-express basis for interpretation, including: matters of logic with respect to arrangement of steps or operational flow; plain meaning derived from grammatical organization or punctuation; the number or type of configurations described in the specification. [0085] It will be apparent to those skilled in the art that various modifications and variations may be made without departing from the scope or spirit. Other configurations will be apparent to those skilled in the art from consideration of the specification and practice described herein. It is intended that the specification and described configurations be considered as exemplary only, with a true scope and spirit being indicated by the following claims.

Claims

1. A method comprising: receiving, by a first computing device, a plurality of content requests associated with a plurality of client devices; for each content request of the plurality of content requests, determining, based on at least one portion of content associated with the content request being unavailable, at least one prioritization parameter associated with the content request; sending, to a second computing device, a multiplexed request associated with the plurality of content requests, wherein the second computing device determines a prioritization schedule based on the multiplexed request, and wherein the multiplexed request comprises the at least one prioritization parameter for each content request of the plurality of content requests; receiving, from the second computing device based on the prioritization schedule, the at least one portion of content for each content request of the plurality of content requests; and for each client device of the plurality of client devices, causing the at least one portion of content associated with the content request corresponding to the client device to be output.
2. The method of claim 1, wherein the at least one prioritization parameter associated with each content request, of the plurality of content requests, is based on one or more of: a buffer length, a buffer size, an amount of time, or an amount of memory associated with a buffer of the client device corresponding to that content request.
3. The method of claim 1, wherein the at least one prioritization parameter associated with each content request, of the plurality of content requests, is based on a content type corresponding to that content request, and wherein the content type comprises one or more of: low-latency live streaming content, low-delay streaming content, linear content, on-demand content, high-concurrency content, high-priority content, or low-priority content.
4. The method of claim 1, wherein the at least one prioritization parameter associated with each content request, of the plurality of content requests, is based on a request type corresponding to that content request, and wherein the request type comprises one or more of: a high-priority request, a low-priority request, a manifest request, a request for at least one initialization segment, a request for at least one partial segment, a request for at least one representation associated with low-delay streaming, or a request for at least one enhancement layer segment.
5. The method of claim 1, wherein receiving, from the second computing device, the at least one portion of content for each content request of the plurality of content requests comprises: determining, by the second computing device, based on the multiplexed request and the at least one prioritization parameter for each content request of the plurality of content requests, the prioritization schedule; and sending, based on the prioritization schedule, a plurality of responses to the first computing device.
6. The method of claim 5, wherein at least one of: the second computing device responds to each content request, of the plurality of content requests, based on the multiplexed request and the

prioritization schedule; or each response of the plurality of responses comprises the at least one portion of content.

7. The method of claim 1, wherein the at least one prioritization parameter comprises an urgency parameter or an incremental parameter, and wherein the multiplexed request comprises the urgency parameter or the incremental parameter for each content request of the plurality of content requests.

8. A method comprising: receiving, from a first computing device, a multiplexed request associated with a plurality of content requests, wherein the multiplexed request comprises, for each content request of the plurality of content requests, at least one prioritization parameter, and wherein the plurality of content requests are each associated with at least one portion of content; determining, based on the at least one prioritization parameter for each content request of the plurality of content requests, a prioritization schedule associated with the plurality of content requests; and for each content request of the plurality of content requests, and based on the prioritization schedule, sending a response to the first computing device, wherein the response causes the at least one portion of content associated with the content request to be output.

9. The method of claim 8, wherein the prioritization schedule indicates, for each content request of the plurality of content requests, at least one of: a time or an order associated with sending the response to the first computing device.

10. The method of claim 8, wherein the response for each content request, of the plurality of content requests, comprises at least one portion of: a frame, a chunk, a segment, a manifest file, or a representation element associated with the at least one portion of content associated with the content request.

11. The method of claim 8, wherein receiving the multiplexed request comprises: receiving, by the first computing device, the plurality of content requests; and for each content request of the plurality of content requests, determining, by the first device, that the at least one portion of content associated with the content request is unavailable.

12. The method of claim 8, further comprising determining, for a first content request of the plurality of content requests, and based on at least one portion of content associated with the first content request, at least one modification to the prioritization schedule, wherein the at least one modification is associated with a time or an order associated with sending the response associated with the at least one portion of content to the first computing device.

13. The method of claim 8, wherein the at least one prioritization parameter, for each content request of the plurality of content requests, comprises an urgency parameter or an incremental parameter, and wherein the multiplexed request comprises the urgency parameter or the incremental parameter for each content request of the plurality of content requests.

14. The method of claim 8, wherein: each content request, of the plurality of content requests, is indicative of at least one of: a buffer status associated with the client device, a request type, or a content type associated with the at least one portion of content; the buffer status is indicative of one or more of: a buffer length, a buffer size, an amount of time, or an amount of memory; the content type comprises one or more of: low-latency live streaming content, low-delay streaming content, linear content, on-demand content, high-concurrency content, high-priority content, or low-priority content; and the request type comprises one or more of: a high-priority request, a low-priority request, a manifest request, a request for at least one initialization segment, a request for at least one partial segment, a request for at least one representation associated with low-delay streaming, or a request for at least one enhancement layer segment.

15. A method comprising: sending, to a first computing device, a first multiplexed request associated with a subset of a plurality of content requests, wherein the first multiplexed request comprises at least one prioritization parameter for each content request of the subset; sending, to a second computing device, a second multiplexed request associated with a remainder of the plurality of content requests, wherein the second multiplexed request comprises at least one prioritization parameter for each content request of the remainder; causing, based on at least one portion of

content for each content request of the subset received via the first computing device according to a first prioritization schedule, the at least one portion of content for each content request of the subset to be output, wherein the first prioritization schedule is based on the at least one prioritization parameter for each content request of the subset; and causing, based on at least one portion of content for each content request of the remainder received via the second computing device according to a second prioritization schedule, the at least one portion of content for each content request of the subset to be output by a remainder of the plurality of client devices, wherein the second prioritization schedule is based on the at least one prioritization parameter for each content request of the remainder.

16. The method of claim 15, further comprising, for each content request of the subset, determining, based on the at least one portion of content associated with the content request being unavailable, the at least one prioritization parameter for the content request.

17. The method of claim 15, further comprising, for each content request of the remainder, determining, based on the at least one portion of content associated with the content request being unavailable, the at least one prioritization parameter for the content request.

18. The method of claim 15, wherein each content request, of the plurality of content requests, is indicative of one or more of: a buffer length, a buffer size, an amount of buffer time, or an amount of buffer memory associated with a client device corresponding to that content request, and wherein the at least one prioritization parameter for that content request is based on one or more of: the buffer length, the buffer size, the amount of buffer time, or the amount of buffer memory associated with the client device.

19. The method of claim 15, wherein each content request, of the plurality of content requests, is indicative of a content type corresponding to that content request, wherein the at least one prioritization parameter for that content request is based on the content type, and wherein the content type comprises one or more of: low-latency live streaming content, low-delay streaming content, linear content, on-demand content, high-concurrency content, high-priority content, or low-priority content.

20. The method of claim 15, wherein each content request, of the plurality of content requests, is indicative of a request type corresponding to that content request, wherein the at least one prioritization parameter for that content request is based on the request type, and wherein the request type comprises one or more of: a high-priority request, a low-priority request, a manifest request, a request for at least one initialization segment, a request for at least one partial segment, a request for at least one representation associated with low-delay streaming, or a request for at least one enhancement layer segment.
